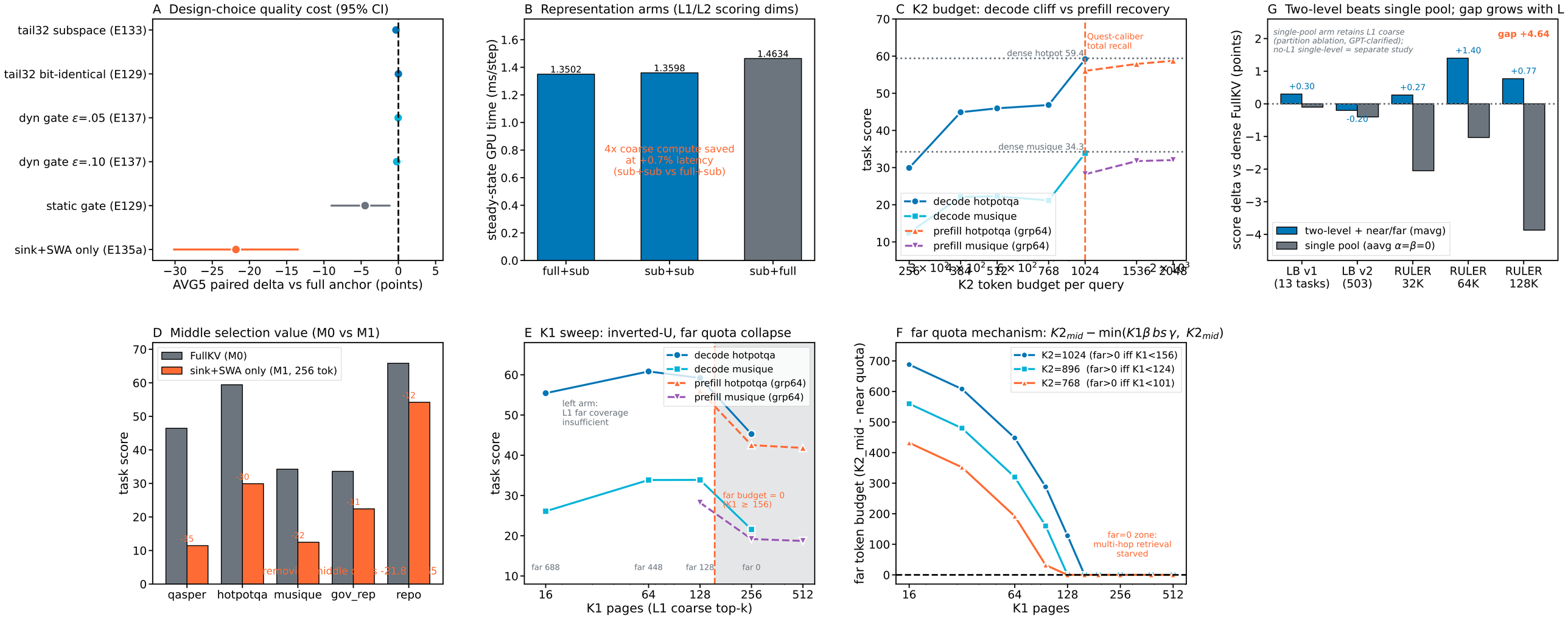
A Design-choice quality cost (95% CI)
tail32 subspace (E133)
tail32 bit-identical (E129)
dyn gate ε=.05 (E137)
dyn gate ε=.10 (E137)
static gate (E129)
sink+SWA only (E135a)
−30 −25 −20 −15 −10 −5 0 5
AVG5 paired delta vs full anchor (points)
B Representation arms (L1/L2 scoring dims)
steady-state GPU time (ms/step)
1.3502
1.3598
1.4634
4x coarse compute saved
at +0.7% latency
(sub+sub vs full+sub)
full+sub
sub+sub
sub+full
C K2 budget: decode cliff vs prefill recovery
task score
dense hotpot 59.4
dense musique 34.3
Quest-caliber
total recall
decode hotpotqa
decode musique
prefill hotpotqa (grp64)
prefill musique (grp64)
256 384 512 768 1024 1536 2048
K2 token budget per query
G Two-level beats single pool; gap grows with L
single-pool arm retains L1 coarse
(partition ablation, GPT-clarified);
no-L1 single-level = separate study
gap +4.64
+0.30
-0.20
+0.27
+1.40
+0.77
score delta vs dense FullKV (points)
two-level + near/far (mavg)
single pool (aavg α=β=0)
LB v1
(13 tasks)
LB v2
(503)
RULER
32K
RULER
64K
RULER
128K
D Middle selection value (M0 vs M1)
FullKV (M0)
sink+SWA only (M1, 256 tok)
task score
-35
-30
-22
-11
-42
qasper
hotpotqa
musique
gov_rep
repo
E K1 sweep: inverted-U, far quota collapse
decode hotpotqa
decode musique
prefill hotpotqa (grp64)
prefill musique (grp64)
left arm:
L1 far coverage
insufficient
far budget = 0
(K1 ≥ 156)
far 688
far 448
far 128
far 0
task score
16 64 128 256 512
K1 pages (L1 coarse top-k)
F far quota mechanism: K2mid − min(K1βbsγ, K2mid)
K2=1024 (far>0 iff K1<156)
K2=896 (far>0 iff K1<124)
K2=768 (far>0 iff K1<101)
far=0 zone:
multi-hop retrieval
starved
far token budget (K2_mid - near quota)
16 64 128 256 512
K1 pages